



# UNIVERSITY OF PLYMOUTH

## **NSBM GREEN UNIVERSITY**

Faculty of Computing

### **PUSL2022 Introduction to IOT**

Internet Of Things(IOT) Group Project

#### **IOT-Powered Smart Thermostat**

Group No: Group BR

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## Table of Contents

|                                 |          |
|---------------------------------|----------|
| <b>Introduction .....</b>       | <b>3</b> |
| <b>Objectives .....</b>         | <b>4</b> |
| <b>Background Study .....</b>   | <b>5</b> |
| <b>Mapping Parameters .....</b> | <b>6</b> |
| Innovation .....                | 6        |
| Feasibility.....                | 6        |
| Sustainability .....            | 6        |
| Scalability .....               | 6        |
| Market Demand.....              | 6        |
| Usability.....                  | 7        |
| <b>Outcome .....</b>            | <b>7</b> |
| <b>Conclusion .....</b>         | <b>8</b> |

## Introduction

Heating and cooling their homes accounts for almost 40% of global energy consumption, with households being a major contributor to global greenhouse gas emissions (as well). Conventional thermostats typically require inefficient use of energy through basic manual controls or inflexible programming that doesn't change based on real time. This may lead to an unnecessary increase in energy consumption, higher bills and also creating an environmental problem.

Automatically, these thermostats don't learn about a person's habits or how the home is changing due to different environmental conditions, resulting in overheated or underheated homes even when unoccupied. Our project tackles this issue through development of a smart IoT powered thermostat based on accessing real time data from advanced environmental sensors integrated to optimize energy consumption. The thermostat is able to monitor indoor temperature, humidity and occupancy until they are continuously monitored and passed to it in order to make informed automated adjustments so that user comfort is optimal while energy is wasted to the lowest level.

This novel solution corresponds to the Sustainable Development Goals adopted by the United Nations. In particular, it moves energy efficiency and promotes clean energy technologies in support of SDG 7 (Affordable and Clean Energy). Also, it is addressing SDG 12 (Responsible consumption and production) related to sustainable energy consumption practices and reducing waste. The last is that the thermostat also contributes to SDG 13 (Climate Action) by significantly reducing carbon emissions caused from residential energy use.

The proposed thermostat, thanks to the use of cutting-edge technology, not only improves user comfort but also works as a vital tool to fight against climate change and minimize the usage of energy. This system demonstrates the power of smart home tech through real time data collection and automated adjustment of behaviors, supporting users to live more sustainably and make decisions that favor both their household and our planet.

## Objectives

The smart thermostat project is an innovative, IoT powered solution for automating and optimizing temperature control in homes. The main objectives include:

1. Automate Temperature Control: Data from environmental sensors (temperature, humidity, occupancy) will feed the thermostat, and it will automatically adjust heating and cooling systems to save users from having to play around with them manually and save energy. This objective assists in realization of SDG 7 (Affordable and Clean Energy) by making clean energy technology affordable.
2. Enable Remote Control via Mobile Application: Users will be able to control the thermostat remotely, see hours and days of energy usage, and get environment notifications via a mobile app. This capability enables responsible energy management, enabling SDG 12 (Responsible Protection) by enabling efficient energy use and the insights to minimize consumption.
3. Optimize Energy Consumption: The system is integrated with occupancy sensors, which allow the temperature to be set only when the space is unoccupied to avoid wasted energy. The provision of this feature is in line with SDG 12 commitment to the responsible use of the energy.
4. Integrate Environmental Sensors: It watches indoor conditions all the time, fine-tuning the comfort while using the least possible amount of energy. SDG 7 is supported through this objective by forcing the use of energy that's both efficient and tailored to real time conditions.
5. Provide Energy Usage Insights: The prevailing energy usage data for which will be accessible through historical data in the mobile application to increase user awareness and encourage sustainable ways. Guiding users to reduce their environmental footprint is supported by SDG 12 and SDG 13 (Climate Action).
6. Promote Sustainability: That will enable efficient operation of HVAC systems, lowering energy bills and greenhouse gas emissions, what it is basically about climate action.
7. Reduce Carbon Footprint: The project works to lower the household carbon footprint by optimising energy use, supporting SDG13 by reducing the need for electricity powered by fossil fuels.

Overall, these objectives serve simultaneously to highlight the practical yet substantial benefits to the user from the smart thermostat, efforts to make user comfortable, provide energy savings and pose substantial impact on the environment.

## **Background Study**

Household heating and cooling systems are among the largest consumers of energy and, as a result, they account for a great deal of global energy consumption and greenhouse gas emissions. Nearly 40 percent of the world's energy goes to buildings, and heating, ventilation and air conditioning (HVAC) systems are leading offenders, according to the International Energy Agency (IEA). These systems can result in a great deal of energy waste because of inefficiencies in the regulating of them, worsening the environmental problems and making consumers pay more for their energy.

With current smart thermostat solutions like Google Nest and Ecobee making strides toward addressing any inefficiencies with basic automation features, this is where temperature is at its most challenging. Many of the existing models, however, cannot optimize on energy consumption sufficiently. Most often, these sorts of systems do not adapt on the fly to environmental variations, for example temperature, humidity, or occupancy patterns; rather they are based upon time schedules or simple algorithms and do not change dynamically over time.

Using Internet of Things (IoT) technology, this project aims to increase the capabilities of smart thermostats. Our approach consists of a paradigm shift from being a passive user and user of a health monitoring system to an active participant in developing a network of advanced sensors that can monitor indoor conditions. Sensors on these devices will monitor values like temperature, humidity and occupancy levels before the thermostat can make intelligent real time adjustments for HVAC systems. Through automation of these processes, our smart thermostat can drastically reduce unnecessary energy consumption, saving both homeowners money and reducing the environmental impact.

The project aligns with several Sustainable Development Goals (SDGs): it helps to achieve SDG 7 (Affordable and Clean Energy), by promoting energy efficiency; encourages responsible energy consumption in line with SDG 12; and contributes to climate action by reducing greenhouse gas emissions in accordance with SDG 13. This project focuses on how real time energy usage optimization can offer a more efficient and sustainable solution to modern households in need of innovation in energy management.

## **Mapping Parameters**

### **Innovation**

Incorporating real time environmental data from next generation sensors, the proposed smart thermostat is a great innovator in the field of energy management. Our system is different from the traditional (or existing) smart thermostats: they are mostly programmed based on schedules, but ours is constantly validating the indoor conditions, such as the temperature, humidity, occupancy levels. This capability provides the ability to precisely, automatically adjust HVAC systems reflecting changes in the environment. The thermostat is simply doing its job of ensuring that heating and cooling occurs only, when necessary, while doing so helps increase user comfort and also lowering energy waste.

### **Feasibility**

Our view is that the feasibility of this project is founded in using inexpensive, off the shelf components, such as temperature and humidity sensor, microcontroller, and wireless communication modules. The approach here also guarantees that the system is cost effective and technically viable for broad adoption. Furthermore, mobile app development integration provides remote access & control of the heating and cooling systems, so that users can easily manage their temperatures. However, these two things are critical for encouraging user engagement and maximizing energy use.

### **Sustainability**

In line with Sustainable Development Goals (SDGs), this project relates to SDG 7 (Affordable and Clean Energy). The thermostat helps with access to clean energy technologies through energy consumption optimization and reduced energy bills. In addition, the promotion of energy efficiency as well as the decline in household emissions is also a support of SDG 12 (Responsible consumption and production) and SDG 13 (Climate action).

### **Scalability**

The project has big potential for scalability starting initially with HVAC systems. There are also future integrations like other smart home devices like lighting systems, automated window treatments and energy efficient appliances. This scalability helps accelerating SDG 12: for energy efficiency along a wide spectrum of household systems, towards a more sustainable living environment.

### **Market Demand**

We are very optimistic about our smart thermostat in light of the growing consumer awareness of energy efficient solutions resulting from increased energy prices and growing environmental concerns. This project aims to address an expanding market, SDG 7, a call for affordable and

accessible energy solutions to consumers looking for devices to reduce costs and increase sustainability.

## **Usability**

As such, the thermostat will come with an extremely intuitive mobile app for the user experience. Users in this app will be able to change temperature settings, view current environmental data, and look over energy consumption reports. The app is designed with customizable preferences, convenient of use, integrated with IoT technology while allows remote access of SDG 12 responsibly according to energy consumption.

## **Outcome**

The purpose of this project is to produce a fully working prototype that is a smart thermostat which has built in features to help optimize energy consumption and be convenient for the user. Key functionalities of this innovative system include:

- **Automated Temperature Control:** With the use of real time environmental data from sensors, the thermostat will adjust heating and cooling settings automatically to give you optimal indoor comfort. It removes the necessity of adjusting manually the way that HVAC systems function with the current environmental conditions, while eliminating energy waste.
- **Remote Management via Mobile App:** Users will be able to manage their thermostat settings remotely via a user-friendly mobile application and real time indoor condition monitoring. It allows users to easily adjust the temperature settings and receive notifications about energy usage and system performance so that users can make well educated energy consumption decisions.
- **Energy Optimization Features:** The smart thermostat will calculate when to heat and cool less using occupancy sensors and smart algorithms when the spaces are unoccupied. Thanks to this proactive approach not only will users save money, but overall demand of energy will be reduced, making it more environmentally sustainable.
- **Detailed Energy Consumption Reports:** But the mobile app will give complete reports on energy consumption, and a user will be able to monitor the amount of used energy, and which strategies used to work more efficiently. This contributes SDG 12 (Responsible consumption production) and SDG 7 (Affordable and clean energy).

Through these outcomes, the project will enable a dramatic reduction of household energy consumption and help meet SDG 13 climate action efforts. Also, that smart thermostat addresses comfort and money savers, while it acts as a critical tool to stop the warming of the planet, to make the future more sustainable for all.

## **Conclusion**

This IoT-powered thermostat project addresses a critical challenge in modern households: Heat and cooling systems being inefficient and consuming a lot of energy. The smartest thermostat intelligently optimizes HVAC operations based on real time data from a network of sensors to achieve the right balance of energy efficiency and user comfort. This is an innovative approach which minimizes energy waste, while improving the living environment for the consumers effectively.

The project very suits the Sustainable Development Goals (SDGs), more specifically SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action). It encourages households to follow responsible energy consumption practices by means of promoting energy efficient technologies, that in turn result in reduced utility bills and small environmental footprint.

Additionally, the thermostat is one of many important devices in continuing the fight against climate change by cutting greenhouse gas emissions that stem from using too much energy. In fact, this project demonstrates the great potential of smart home technologies to address the issue of energy and climate to improve a sustainable future. Not only does its successful implementation benefit individual households, but it also serves to help realize global pursuance that seek to promote environmental sustainability and resilience at the world level.